

CANDIDATE'S NAME: _____

CTG: _____

YISHUN JUNIOR COLLEGE
2017 JC2 PRELIMINARY EXAMINATION

CHEMISTRY
HIGHER 2

9729/01

Paper 1 Multiple Choice Questions

FRIDAY 15 SEPTEMBER 2017
0800hrs – 0900hrs
(1 hour)

Additional Materials: Optical Mark Sheet (OMS)
Data Booklet



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, CTG, and NRIC / FIN number on the Optical Mark Sheet (OMS), and shade the corresponding boxes for your NRIC / FIN number.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

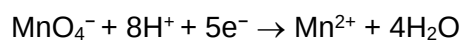
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

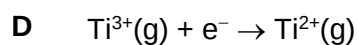
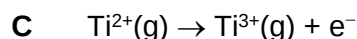
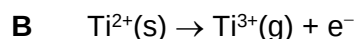
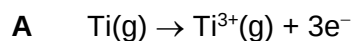
- 1 An element R can exist in a few oxidation states. 0.01 mol of R^{2+} requires 0.004 mol of acidified $KMnO_4$ for complete reaction. The half equation for reduction of MnO_4^- is



What is the final oxidation state of R?

- A** +4 **B** +3 **C** 0 **D** -1

- 2 Which equation corresponds to the third ionisation energy of titanium (Ti)?



- 3 *Use of the Data Booklet is relevant to this question.*

Which of the following particles would, on gaining an electron, have a half-filled set of p orbitals?

- 1 B^- 2 C 3 N 4 N^+

- A** 1, 2 and 3

- B** 1, 2 and 4

- C** 3 only

- D** 2 and 4

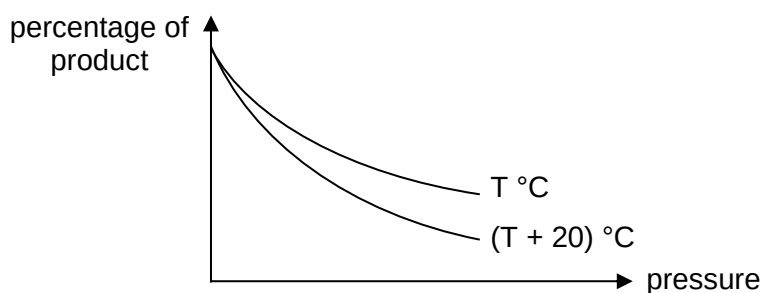
- 4 Which gas shows the greatest deviation from ideality?

- A** CH_4 **B** SO_3 **C** C_2H_2 **D** O_2

- 5 A gas at a pressure of 4.50 atm is heated from 25 °C to 480 °C and simultaneously compressed to one-third of its original volume. What will be the final pressure?

- A** 3.79 atm **B** 34.1 atm **C** 86.4 atm **D** 259 atm

- 6 The graph below shows how the percentage of product present at equilibrium varies with temperature and pressure for a reaction.



Which reaction could the graph represent?

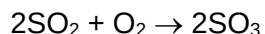
- A** $4\text{Fe(s)} + 3\text{O}_2\text{(g)} \rightleftharpoons 2\text{Fe}_2\text{O}_3\text{(s)}$ $\Delta H = -1644 \text{ kJ mol}^{-1}$
- B** $\text{N}_2\text{O}_4\text{(g)} \rightleftharpoons 2\text{NO}_2\text{(g)}$ $\Delta H = +57 \text{ kJ mol}^{-1}$
- C** $2\text{C(s)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{CO(g)}$ $\Delta H = -222 \text{ kJ mol}^{-1}$
- D** $\text{CO(g)} + \text{Cl}_2\text{(g)} \rightleftharpoons \text{COCl}_2\text{(s)}$ $\Delta H = +86 \text{ kJ mol}^{-1}$
- 7 Values for the ionic product of water, K_w , at two different temperatures are given below.

Temperature / °C	$K_w / \text{mol}^2 \text{dm}^{-6}$
25	1.00×10^{-14}
50	5.48×10^{-14}

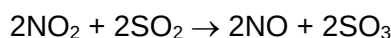
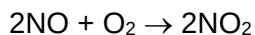
What is correct for pure water at 50 °C?

- A** $[\text{H}^+] > [\text{OH}^-]$
- B** $\text{pH} = 2.34 \times 10^{-7}$
- C** $\text{pH} > 7$
- D** $\text{pH} < 7$
- 8 What is a satisfactory indicator for the titration of 0.10 mol dm^{-3} aqueous methylamine with 0.10 mol dm^{-3} ethanoic acid?
- A** methyl red (pH range 4.2 – 6.3)
- B** bromothymol blue (pH range 6.0 – 7.6)
- C** thymol blue (pH range 8.0 – 9.6)
- D** there is no suitable indicator

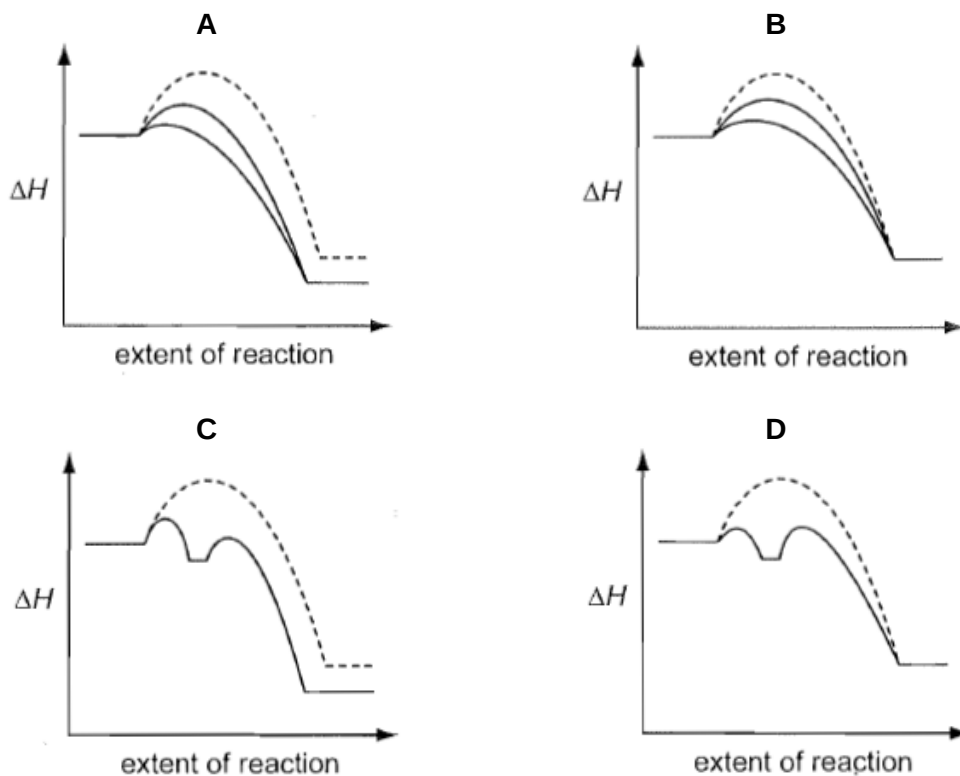
- 9 The uncatalysed reaction between SO_2 and O_2 is slow.



The reaction is speeded up in the presence of the homogeneous catalyst NO, which participates as follows.



Which reaction pathway diagram is most appropriate for describing the enthalpy changes occurring during the catalysed reaction? In each case the reaction pathway for the uncatalysed reaction is shown as a dashed line.



- 10 The enthalpy change of formation of rubidium bromide, RbBr , can be calculated using a Born-Haber cycle.

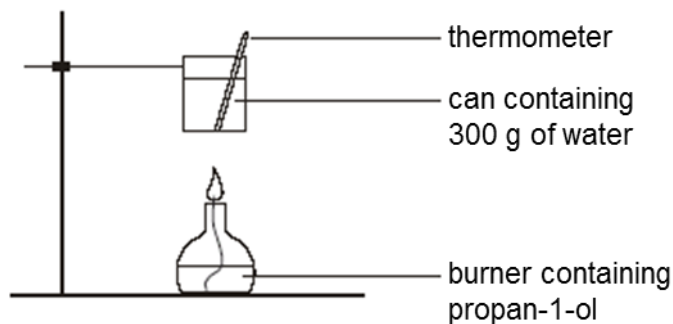
The relevant enthalpy changes are shown in the table.

	enthalpy change / kJ mol^{-1}
$\text{Rb(s)} \rightarrow \text{Rb(g)}$	+81
$\text{Br}_2(\text{l}) \rightarrow 2\text{Br(g)}$	+224
$\text{Rb(g)} \rightarrow \text{Rb}^+(\text{g}) + \text{e}^-$	+403
$\text{Br(g)} + \text{e}^- \rightarrow \text{Br}^-(\text{g})$	-324
$\text{Rb}^+(\text{g}) + \text{Br}^-(\text{g}) \rightarrow \text{RbBr(s)}$	-687

What is the enthalpy change of formation of RbBr ?

- A -303 kJ mol^{-1} B -415 kJ mol^{-1}
C -577 kJ mol^{-1} D -627 kJ mol^{-1}

- 11 A student used the set-up below to heat a can containing 300 g of water.



The following data were recorded:

mass of propan-1-ol burnt $= m$ g
 change in temperature of water $= \Delta T$ °C

Given that:

relative molecular mass of propan-1-ol $= 60.0$
 enthalpy change of combustion of propan-1-ol $= -2021$ kJ mol⁻¹
 specific heat capacity of water $= c$ J g⁻¹ K⁻¹

What is the efficiency of this heating process?

A $\frac{m \times 2021 \times 1000}{300 \times c \times \Delta T \times 60.0} \times 100\%$

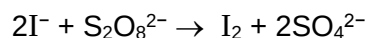
B $\frac{m \times c \times \Delta T \times 60.0}{300 \times 2021 \times 1000} \times 100\%$

C $\frac{300 \times c \times \Delta T \times 60.0}{m \times 2021} \times 100\%$

D $\frac{300 \times c \times \Delta T \times 60.0}{m \times 2021 \times 1000} \times 100\%$

- 12 Use of the Data Booklet is relevant to this question.

Peroxodisulfate(VI), S₂O₈²⁻, is capable of oxidising iodide as shown in the following equation.



Which metal ion can be a suitable catalyst for this reaction?

- 1 Cu²⁺ 2 Co²⁺ 3 Mn²⁺ 4 V²⁺

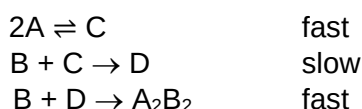
A 1, 2 and 3

B 2 and 3

C 2 only

D 4 only

- 13 Consider the reaction $2A + 2B \rightarrow A_2B_2$.
The mechanism involves the following steps:



Based on the information, what is the rate equation for this reaction?

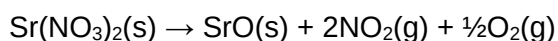
- A rate = $k [A]^2 [B]^2$
 B rate = $k [B] [C]$
 C rate = $k [A]^2 [B]$
 D rate = $k [A] [B]^2$
- 14 Element J is in Period 3 of the Periodic Table. The following four statements were made about the properties of element J or its compounds.

Three statements are correct descriptions and one is false.

Which statement does **not** fit with the other three?

- A Adding NaOH(aq) to the solution resulting from the reaction of a chloride of J with water produces a white precipitate which is soluble in an excess of NaOH(aq).
 B Element J is a solid at room temperature.
 C Element J has the highest melting point among the period 3 elements.
 D The oxide of element J is soluble in hydrochloric acid.
- 15 A student was given a sample of strontium carbonate that was mixed with strontium nitrate. He carried out two separate experiments using the same mass of the solid mixture.

Strontium nitrate decomposes according to the following equation.



Experiment 1: The sample was heated strongly to constant mass and the volume of gas evolved was found to be 200 cm³.

Experiment 2: Excess dilute sulfuric acid was added to the sample and the volume of gas evolved was found to be 75 cm³.

All volumes were measured at the same temperature and pressure.

What is the mole ratio of strontium carbonate to strontium nitrate in the sample?

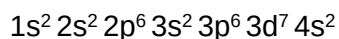
- A 3 : 2 B 3 : 5 C 6 : 5 D 8 : 5

- 16 Use of the Data Booklet is relevant to this question.

By using the relevant $E^\ominus$ values, what are the vanadium containing products formed when chlorine is passed through a solution that contains $\text{VO}^{2+}(\text{aq})$?

- A V^{3+}
- B VO_2^+
- C VO_2^+ , VO_3^- and V^{3+}
- D VO_2^+ and VO_3^-

- 17 Element Y has the following electronic configuration.



Which compounds do **not** exist?

- 1 YO_2Cl_2
- 2 $[\text{Y}(\text{C}_2\text{O}_4)_2(\text{H}_2\text{O})_2]\text{Cl}$
- 3 $\text{Na}_2[\text{Y}(\text{CN})_6]$
- 4 YCl_2

- A 1 only
- B 1, 2 and 3
- C 2 and 3
- D 2, 3 and 4

- 18 Platinum(IV) chloride is combined with ammonia to form a single product. The product is a different platinum(IV) compound which contains a cation with a 2+ charge and has a co-ordination number of 6.

What is the formula of this product?

- A $\text{Pt}(\text{NH}_3)_6\text{Cl}_2$
- B $\text{Pt}(\text{NH}_3)_3\text{Cl}_4$
- C $\text{Pt}(\text{NH}_3)_4\text{Cl}_4$
- D $\text{Pt}(\text{NH}_3)_6\text{Cl}_4$

- 19 The C–H bond lengths of three hydrocarbons are given in the table below.

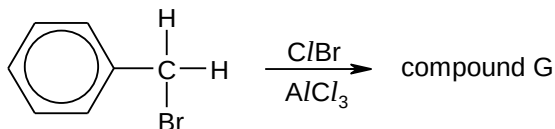
compound	structural formula	C–H bond length / nm
ethane	CH_3CH_3	0.110
ethene	$\text{CH}_2=\text{CH}_2$	0.108
ethyne	$\text{CH}\equiv\text{CH}$	0.106

Which statement best explains the shortest C–H bond length observed in ethyne?

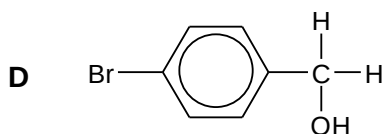
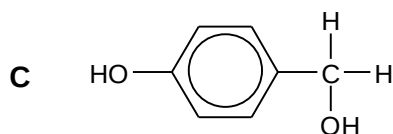
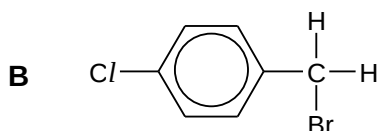
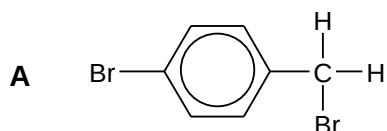
- A** The carbon-carbon triple bond in ethyne has the greatest bond energy.
- B** The bond angle around the carbon atom in ethyne is the largest.
- C** The orbital that carbon used in formation of the C–H bond in ethyne has the greatest s orbital character.
- D** A sp-sp overlap occurs between the two carbon atoms in ethyne.
- 20 Which statements best describe the reaction mechanism involved in the conversion of chloroethane to ethanol?
- 1 The reaction mechanism involves a carbocation.
 - 2 The reaction mechanism involves nucleophilic attack of OH^- in an overall second order reaction.
 - 3 The reaction mechanism involves heterolytic fission of the C–Cl bond, followed by a reaction with OH^- .
 - 4 The activation energy of the first step of the mechanism is greater than the activation energy of the second step of the mechanism.
- A** 1 and 3 only
- B** 2 only
- C** 1, 3 and 4 only
- D** 2 and 4 only
- 21 How many constitutional isomers can $\text{C}_4\text{H}_9\text{Cl}$ form?

- A** 3 **B** 4 **C** 5 **D** 6

- 22 Bromine, in the presence of anhydrous aluminium chloride, undergoes a substitution reaction with benzene forming bromobenzene. When bromomethylbenzene reacts with ClBr , in the presence of anhydrous aluminium chloride, a similar reaction occurs and compound G is formed.

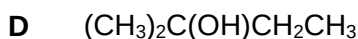
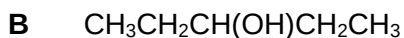
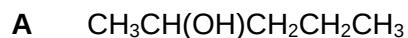


What could compound G be?



- 23 An alcohol D has the molecular formula of $\text{C}_5\text{H}_{11}\text{OH}$. On heating D with concentrated sulfuric acid, two products E and F are formed, E being the minor product. On heating E with hot acidified potassium manganate(VII), one of the products gives a pale yellow precipitate with alkaline aqueous iodine.

What could D be?



- 24 Which of the following reagents **cannot** be used to distinguish between ethanol and propanone?

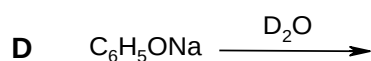
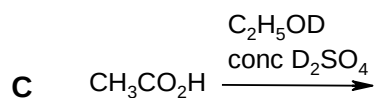
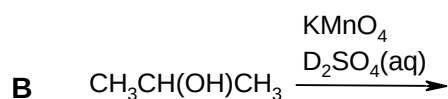
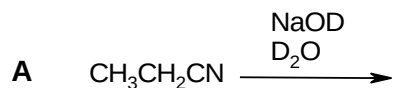
A alkaline aqueous iodine

B acidified potassium dichromate(VI)

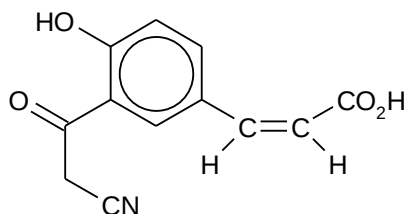
C 2,4-dinitrophenylhydrazine

D sodium metal

25 Which reaction yields a carbon compound containing deuterium, D? [D = ^2H]



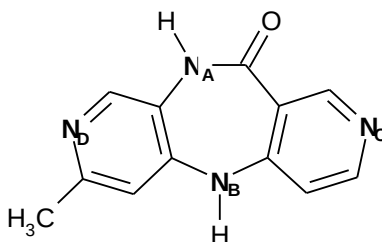
26 Which statements about compound P are correct?



compound P

- 1 4 moles of hydrogen gas will react completely with 1 mole of compound P in the presence of platinum metal.
 - 2 It will exist as a pair of stereoisomers.
 - 3 It reacts with ethanoic acid in the presence of concentrated sulfuric acid to form a sweet-smelling product.
- A 1, 2 and 3
- B 2 only
- C 1 and 2 only
- D 2 and 3 only

- 27 An analogue of Nevirapine which is a synthetic antiviral drug is shown below.



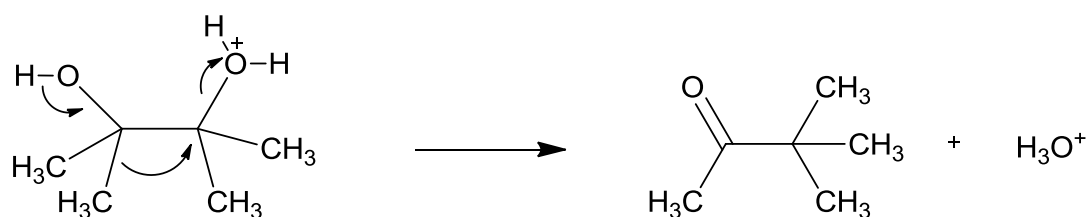
The four nitrogen atoms are labelled as N_A , N_B , N_C and N_D .

N_C and N_D are part of the pyridine rings where the lone pair of electrons on the nitrogen atoms are not delocalised into the pyridine rings.

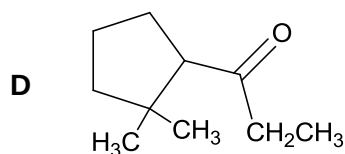
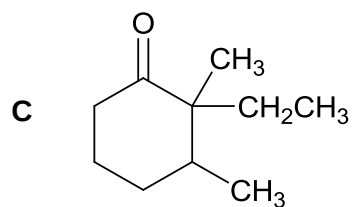
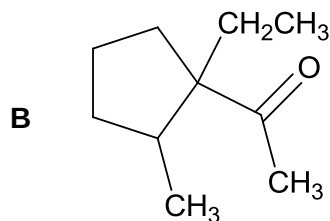
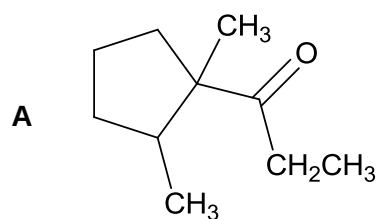
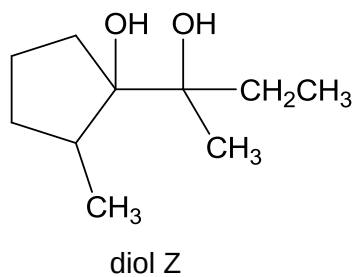
Which sequence shows the four nitrogen atoms in decreasing order of basicity?

- A $N_B > N_D > N_C > N_A$
 - B $N_D > N_C > N_A > N_B$
 - C $N_C > N_D > N_B > N_A$
 - D $N_D > N_C > N_B > N_A$
- 28 Which statement about 2-aminopropanoic acid extracted from silkworms is correct?
- A It is able to rotate plane polarised light.
 - B It reacts with ethanoic acid to form an amide.
 - C pH of its aqueous solution drops drastically when a small amount of hydrochloric acid is added.
 - D It decolourises orange aqueous bromine.

- 29 When a 1, 2-diol is treated with a dilute acid, the protonated diol undergoes pinacol rearrangement as shown.

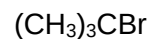
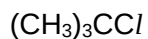
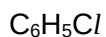
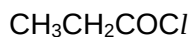


Which of the following products is **not** formed via pinacol rearrangement by adding dilute acid to diol Z?



30 *Use of the Data Booklet is relevant to this question.*

An experiment was conducted to compare the rate of hydrolysis of four halogen-containing compounds. 1 mole of the 4 compounds K, L, M and N are each heated separately with excess NaOH(aq) for 30 seconds and then acidified with HNO₃(aq). Excess AgNO₃(aq) is added and the precipitate formed is filtered, dried and weighed.



K

L

M

N

71.1 g of the precipitate is obtained from compound M. What are the possible masses of the precipitates obtained from the other three halogen-containing compounds?

	mass of precipitate from K / g	mass of precipitate from L / g	mass of precipitate from N / g
A	128.1	0	112.9
B	120.2	0	60.7
C	126.2	40.2	112.9
D	65.2	0	57.3

~ END OF PAPER ~